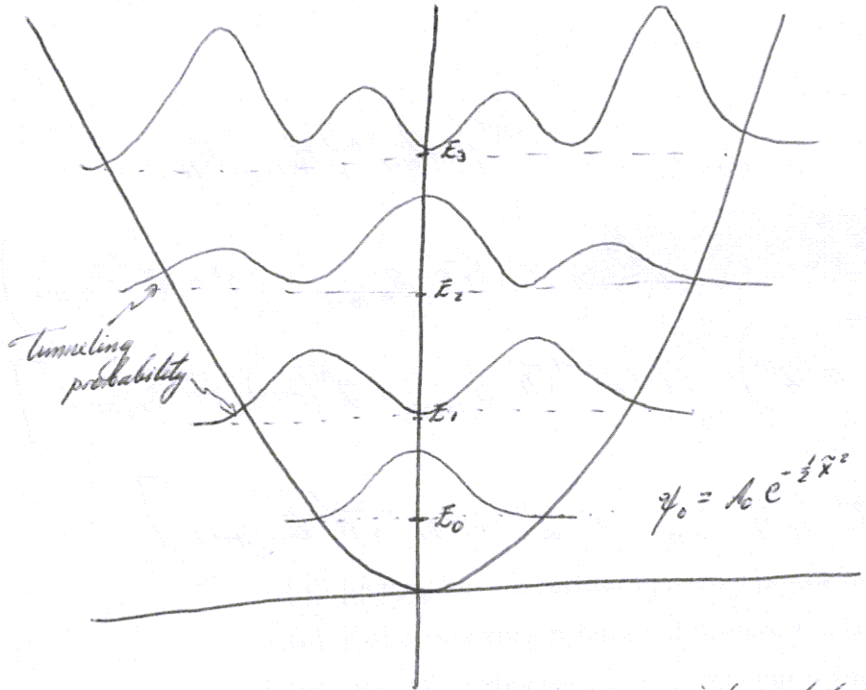
~ Selection rules in the harmonic oscillator



$$P_{\psi_3} = P_{\psi_0} \frac{H_3^2}{48}$$

$$P_{\psi_2} = P_{\psi_0} \frac{H_2^2}{8}$$

$$P_{\psi_1} = P_{\psi_0} \frac{H_1^2}{2}$$

$$\psi_0 = A_0 e^{-\frac{1}{2}\tilde{x}^2} \rightsquigarrow P_{\psi_0} = A_0^2 e^{-\tilde{x}^2}$$

If the harmonic oscillator is interpreted as an atomic system, we expect a probability that an electron with energy E_n undergoes a transition to a state with $E_m < E_n$. m cannot be any integer, but is governed by the selection rules, which indicate which transitions are <u>allowed</u> and which ones are <u>forbidden</u>.

We are concerned with transitions whose associated radiation would be that produced by a dipole antenna lying on the x-axis. Of course, other kinds of radiation (e.g. quadrupolar, magnetic, ...) can occur, but these are less probable. For these reasons, we consider the electric dipole moment

$$d = e x$$

associated to an atom with an electron. The amplitude of a transition determines the likelihood of a change in the electric dipole moment per time unit; this can be computed as (see §9. De la Peña)

$$M_{nm} = \frac{4\omega^3}{3\hbar c^3} |\langle m | d | n \rangle|^2$$

If $M_{nm} = 0 \rightsquigarrow$ forbidden
$M_{nm} \neq 0 \rightsquigarrow$ allowed

As W_{mn} represents the transition prob. per time unit, the mean lifetime of the state with E_n going to E_m is

$$W_{nm}^{-1} .$$

In the current case, we need (recall $d\tilde{x} = \sqrt{\frac{m\omega}{\hbar}}\, dx$)

$$\langle m|\hat{x}|n\rangle = \int dx \int dx' \, \langle m|x\rangle\langle x|\hat{x}|x'\rangle\langle x'|n\rangle$$

$$= \int dx \int dx' \, \psi_m^*(x)\, x\, \delta(x-x')\, \psi_n(x') = \int dx \, \psi_m^*(x)\, x\, \psi_n(x)$$

$$= \sqrt{\frac{\hbar}{m\omega}} \int d\tilde{x} \, \psi_m^*(\tilde{x}) \sqrt{\frac{\hbar}{m\omega}}\, \tilde{x}\, \psi_n(\tilde{x})$$

$$= \frac{\hbar}{m\omega} A_m \int d\tilde{x} \, e^{-\frac{1}{2}\tilde{x}^2}\, \tilde{x}\, H_m(\tilde{x})\, \psi_n(\tilde{x})$$

Use
$$\boxed{\begin{aligned} &\tilde{x} H_n(\tilde{x}) = n\, H_{n-1}(\tilde{x}) + \frac{1}{2} H_{n+1}(\tilde{x}) \\ \& \quad &\int d\tilde{x} \, e^{-x^2} H_n H_m = 2^n n! \sqrt{\pi}\, \delta_{n,m} \end{aligned}}$$

$$\Leftrightarrow \langle m|\hat{x}|n\rangle = \frac{\hbar}{m\omega} A_n A_m \int d\hat{x} \, e^{-\tilde{x}^2} \left(m H_{m-1}(\tilde{x}) + \frac{1}{2} H_{m+1}(\tilde{x}) \right) H_n(\tilde{x})$$

$$= \frac{\hbar}{m\omega} A_n A_m \left(m\, 2^{m-1}(m-1)! \sqrt{\pi}\, \delta_{n,m-1} + \frac{1}{2} 2^{m+1}(m+1)! \sqrt{\pi}\, \delta_{n,m+1} \right)$$

$$= \frac{\hbar}{m\omega} \left(A_{m-1} A_m\, m\, 2^{m-1}(m-1)! \sqrt{\pi}\, \delta_{n,m-1} + \frac{1}{2} A_{m+1} A_m\, 2^{m+1}(m+1)! \sqrt{\pi}\, \delta_{n,m+1} \right)$$

$$A_n = \left(\sqrt{\frac{\hbar\pi}{m\omega}}\, n!\, 2^n \right)^{-1/2}$$

$$\Leftrightarrow \langle m|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{m\omega}} \left[\frac{\sqrt{\pi}}{\sqrt{\pi}} \right] \left[\left((m-1)!\, 2^{m-1} m!\, 2^m\right)^{-1/2} m\, (m-1)!\, 2^{m-1} \delta_{n,m-1} + \left((m+1)!\, 2^{m+1} m!\, 2^m\right)^{-1/2} \times \frac{1}{2} 2^{m+1} (m+1)!\, \delta_{n,m+1} \right]$$

$$\cancel{= \sqrt{\frac{\hbar}{m\omega}} \left[\frac{m A_m}{A_{m-1}} \underbrace{\delta_{n,m-1}}_{m=n+1} + \frac{1}{2} \frac{A_m}{A_{m+1}} \underbrace{\delta_{n,m+1}}_{m=n-1} \right]}$$

$$\Leftrightarrow \quad \langle m|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{m\omega}}\left[m\sqrt{\frac{(m-1)!\,2^{m-1}}{m!\,2^m}}\,\delta_{n,m-1} + \frac{1}{2}\sqrt{\frac{(m+1)!\,2^{m+1}}{m!\,2^m}}\,\delta_{n,m+1}\right]$$

$$= \sqrt{\underset{\overset{?}{m\omega}}{\frac{\hbar}{m\omega}}}\left[m\sqrt{\frac{1}{2m}}\,\delta_{n,m-1} + \frac{1}{2}\sqrt{(m+1)\,2}\,\delta_{n,m+1}\right]$$

$$= \sqrt{\frac{\hbar}{m\omega}}\left[\sqrt{\frac{m}{2}}\,\delta_{n,m-1} + \sqrt{\frac{m+1}{2}}\,\delta_{n,m+1}\right]$$

$$= \sqrt{\frac{\hbar}{m\omega}}\left[\sqrt{\frac{n+1}{2}}\,\delta_{m,n+1} + \sqrt{\frac{n}{2}}\,\delta_{m,n-1}\right]$$

$$\Rightarrow \quad \boxed{\langle n+1|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{m\omega}}\sqrt{\frac{n+1}{2}}\,, \qquad \langle n-1|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{m\omega}}\sqrt{\frac{n}{2}} \qquad \text{allowed!}}$$

$$\& \qquad \langle m|\hat{x}|n\rangle = 0 \quad \forall\ m \neq n\pm 1 \qquad \text{forbidden}$$

$$\Rightarrow \quad \mathcal{M}_{n,n+1} = \frac{4\hbar e^2\omega^3}{3\hbar c^3 m\omega}\,\frac{n+1}{2} \qquad \mathcal{M}_{n,n-1} = \frac{4\hbar e^2\omega^3}{3\hbar c^3 m\omega}\,\frac{n}{2}$$

$$= \frac{2e^2\omega}{3\hbar c^3 m}\left(E_{n+1} - E_0\right) \qquad = \frac{2e^2\omega}{3\hbar c^3 m}\left(E_n - E_0\right)$$

Clearly, there's no transition $E_0 \to E_0$

The lifetime of a state $|n\rangle$ going to $|n-1\rangle$ is the inverse of $\mathcal{M}$. Thus, it reduces as $E_n - E_0$ grows.

OBS. Classically, one can compute the power

$$W = \frac{2e^2\omega^2}{3mc^3}E$$

which quantically is given by

$$W = \hbar\omega\,\mathcal{M}$$

For transition $|n\rangle \to |n-1\rangle$ we find

$$W = \frac{2e^2\omega^2}{3mc^3}\left(E_n - E_0\right)$$

which reduces to the classical limit when $E_n \gg E_0$, i.e $n \gg 0$, as it should be according to the principles of quantum mechanics.

~6.2. The hydrogen atom

~ Two-body problem

The hydrogen atom is composed of a proton and an electron. Due to the mass difference ($m_p \sim 1900\, m_e$), usually one pictures (classically) the electron orbiting around the proton. This classical picture is partly wrong because it's not possible to measure such behavior. However, as in the Bohr atom, this picture allows one to write the eqs. which describe the behavior of the quantum system.

Let's first analyze a two-body system whose interaction potential depends only on the relative position of the bodies. The Hamiltonian of the system is then

$$\hat{H} = \hat{H}_1 + \hat{H}_2 + \hat{V}_{int}(\hat{\vec{r}}_1 - \hat{\vec{r}}_2)$$

$$\hat{H}_i = \frac{\hat{\vec{p}}_i^{\,2}}{2m_i} + \cancel{\hat{V}_i(\hat{\vec{r}}_i)}$$ assume no external potential acting on m_i (closed system)

Since $\hat{V}_{int}$ depends only on $\vec{r}_1 - \vec{r}_2$, it suggests that the 2 degrees of freedom can be rearranged in

$\vec{r} = \vec{r}_1 - \vec{r}_2$ relative position

$\vec{R} = \dfrac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$ center of mass

$\Rightarrow$ the gradients in the new vars.

$$\vec{\nabla}_1 = \frac{d\vec{r}}{d\vec{r}_1}\vec{\nabla}_r + \frac{d\vec{R}}{d\vec{r}_1}\vec{\nabla}_R = \vec{\nabla}_r + \frac{m_1}{M}\vec{\nabla}_R \qquad M \equiv m_1 + m_2$$

$$\vec{\nabla}_2 = -\vec{\nabla}_r + \frac{m_2}{M}\vec{\nabla}_R$$

$$\Rightarrow \frac{\vec{\nabla}_1^{\,2}}{m_1} + \frac{\vec{\nabla}_2^{\,2}}{m_2} = \left(\frac{1}{m_1} + \frac{1}{m_2}\right)\vec{\nabla}_r^{\,2} + \frac{1}{M}\vec{\nabla}_R^{\,2} \equiv \frac{1}{m}\vec{\nabla}_r^{\,2} + \frac{1}{M}\vec{\nabla}_R^{\,2}$$

$$m = \left(\frac{1}{m_1} + \frac{1}{m_2}\right)^{-1} = \frac{m_1 m_2}{m_1 + m_2} \qquad \text{reduced mass}$$

In the position rep., the ~~eq.~~ Schrödinger eq. reads

$$\left(-\frac{\hbar^2}{2m_1}\vec{\nabla}_1^2 - \frac{\hbar^2}{2m_2}\vec{\nabla}_2^2 + V_{int}(\vec{r})\right)\Psi(\vec{r}_1) = E\,\Psi(\vec{r}_1) \quad (\vec{r}_1, \vec{r}_2)$$

$$\Leftrightarrow \left(-\frac{\hbar^2}{2m}\vec{\nabla}_r^2 - \frac{\hbar^2}{2M}\vec{\nabla}_R^2 + V_{int}(\vec{r})\right)\Psi(\vec{r}, \vec{R}) = E\,\Psi(\vec{r}, \vec{R})$$

This kind of problems can be solved by <u>separation of variables</u>

ansatz: $\Psi(\vec{r}, \vec{R}) = \psi(\vec{r})\phi(\vec{R})$

$$\Rightarrow \psi(\vec{r})\left(-\frac{\hbar^2}{2M}\vec{\nabla}_R^2\right)\phi(\vec{R}) + \phi(\vec{R})\left(-\frac{\hbar^2}{2m}\vec{\nabla}_r^2 + V_{int}(\vec{r})\right)\psi(\vec{r}) = E\,\psi(\vec{r})\phi(\vec{R})$$

In the center of mass frame $\phi(\vec{R})$ is not dynamic ~~so~~

$$-\psi(\vec{r})\frac{\hbar^2}{2M}\vec{\nabla}_R^2\phi(\vec{R}) = 0 \qquad (\text{i.e. } \hat{p}_R = 0 \ \& \ E_R = 0)$$

$$\rightsquigarrow \phi(\vec{R}) = \text{const}$$

Thus, we're left with

$$\left(\frac{-\hbar^2}{2m}\vec{\nabla}_r^2 + V_{int}(\vec{r})\right)\psi(\vec{r}) = E\,\psi(\vec{r})$$

~ Central field and angular momentum (<u>Pictures</u>)

Main assumption: $V_{int} = V_{int}(r)$, $r = |\vec{r}_1 - \vec{r}_2| = |\vec{r}| \geq 0$ central potential

En esta situación, the system has rotational symmetry & angular momentum is conserved (in time).

To notice this, let us study the time-evolution of a quantum system governed by: $i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle$

Let's compute now the matrix element evolution

$$\frac{d}{dt}\langle\psi'|\hat{A}|\psi\rangle = \langle\psi'|\frac{\partial\hat{A}}{\partial t}|\psi\rangle + \frac{\partial\langle\psi'|}{\partial t}\hat{A}|\psi\rangle + \langle\psi'|\hat{A}\frac{\partial|\psi\rangle}{\partial t}$$

$$\Leftrightarrow \quad \frac{d}{dt}\langle\psi'|\hat{A}|\psi\rangle = \langle\psi'|\frac{\partial\hat{A}}{\partial t}|\psi\rangle - \frac{1}{i\hbar}\langle\psi'|\hat{H}\hat{A}|\psi\rangle + \frac{1}{i\hbar}\langle\psi'|\hat{A}\hat{H}|\psi\rangle$$

$$= \langle\psi'|\frac{\partial\hat{A}}{\partial t}|\psi\rangle + \frac{1}{i\hbar}\langle\psi'|[\hat{A},\hat{H}]|\psi\rangle$$

Let's suppose that $\hat{A}$ doesn't depend explicitly on t

$$\Rightarrow \boxed{i\hbar\frac{d}{dt}\langle\psi'|\hat{A}|\psi\rangle = \langle\psi'|[\hat{A},\hat{H}]|\psi\rangle}$$

Two possible interpretations

a) Schrödinger picture: $|\psi\rangle = |\psi(t)\rangle$ and the time-evolution of the system is given by the Schrödinger equation

$$i\hbar\frac{d|\psi\rangle}{dt} = \hat{H}|\psi\rangle, \quad \hat{A} \text{ does not evolve}$$

b) Heisenberg picture: $|\psi\rangle \neq |\psi(t)\rangle$ and the time-evolution of the system is due to the evolution of $\hat{A}$ ~~with~~

$$i\hbar\frac{d}{dt}\langle\psi'|\hat{A}|\psi\rangle = \langle\psi'|i\hbar\frac{d}{dt}\hat{A}|\psi\rangle = \langle\psi'|[\hat{A},\hat{H}]|\psi\rangle$$

$$\Rightarrow \boxed{i\hbar\frac{d\hat{A}}{dt} = [\hat{A},\hat{H}]} \quad \text{Heisenberg eq. of motion}$$

$|\psi\rangle$ does not evolve

Note e.g. that in $D=1$

$$[\hat{x},\hat{H}] = \frac{1}{2m}[\hat{x},\hat{p}^2] = \frac{1}{2m}[\hat{x},\hat{p}]\hat{p} + \frac{1}{2m}\hat{p}[\hat{x},\hat{p}] = \frac{i\hbar}{m}\hat{p}$$

$$\Rightarrow \quad \frac{d\hat{x}}{dt} = \frac{\hat{p}}{m}$$

(incidentally $\Rightarrow \langle(\Delta\hat{p})^2(\Delta\hat{H})\rangle \geq \frac{\hbar^2}{4m^2}\langle\hat{p}\rangle^2$: $|\psi\rangle$ isn't eigenstate of $\hat{p}$ & $\hat{H}$)

$$[\hat{p},\hat{H}] = [\hat{p},\hat{V}] = -i\hbar\hat{V}' \quad \leftarrow \text{show using } \langle x|\ldots|\psi\rangle$$

$$\Rightarrow \quad \frac{d\hat{p}}{dt} = -\frac{d\hat{V}}{d\hat{x}} \neq 0 \quad \text{if } \hat{V}\neq 0 \text{ \underline{\&} depends on } \hat{x}$$

Interesting case

$$\boxed{\begin{array}{c}\dfrac{d\hat{A}}{dt} = 0 \\ \Leftrightarrow \quad [\hat{A},\hat{H}] = 0\end{array}} \quad \hat{A} \text{ is a conserved quantity}$$

Recall that classically momentum is conserved only when

$$F = -\nabla V = 0$$

which also applies in the quantum systems!

<u>Noether's theorem</u> states that every conserved quantity is associated with a <u>symmetry</u> of the system, i.e. with a transformation operation which leaves the system invariant.

Let's define a symmetry operator as a unitary operator $\hat{\mathcal{T}}$, such that $\xrightarrow{}$ associated to $\hat{A}$ observable

$$\hat{\mathcal{T}}(\alpha) = e^{-i\alpha\hat{A}/\hbar} \qquad \alpha \in \mathbb{R} \text{ parameter of transf.}$$

leaves $\hat{H}$ invariant, i.e.

$$\hat{\mathcal{T}}^{\dagger}(\alpha)\,\hat{H}\,\hat{\mathcal{T}}(\alpha) \overset{!}{=} \hat{H} \qquad (*)$$

By expanding $\hat{\mathcal{T}}$ as a Taylor series

$$\hat{\mathcal{T}}(\alpha) = \hat{1} - \frac{i\alpha}{\hbar}\hat{A} + \frac{1}{2}\left(\frac{i\alpha}{\hbar}\right)^2\hat{A}^2 + \ldots$$

one can prove that the left-hand side of $(*)$ yields

$$\hat{\mathcal{T}}^{\dagger}(\alpha)\hat{H}\hat{\mathcal{T}}(\alpha) = \hat{H} + \left(\frac{i\alpha}{\hbar}\right)[\hat{A},\hat{H}] + \frac{1}{2}\left(\frac{i\alpha}{\hbar}\right)^2[\hat{A},[\hat{A},\hat{H}]] + \frac{1}{3!}\left(\frac{i\alpha}{\hbar}\right)^3[\hat{A},[\hat{A},[\hat{A},\hat{H}]]] + \ldots$$

$$\overset{!}{=} \hat{H}$$

$$\Longleftrightarrow \quad [\hat{A},\hat{H}] = 0 \quad \Rightarrow \quad \frac{d\hat{A}}{dt} = 0$$

The parameter α may also be a non-commuting (called non-Abelian) operator, but we shall just take it as a number.

Rules of thumb:

Conserved quantity	Parameter	Transformation
E	t	Time shift
p	d	spatial shift
L	ϑ	rotation

$\Rightarrow \quad \hat{H} \neq \hat{H}(t)$ time invariance $\rightsquigarrow$ energy conservation

$\hat{H} \neq \hat{H}(\vec{r})$ spatial invariance $\rightsquigarrow$ momentum conservation

$\hat{H} \neq \hat{H}(\vartheta)$ rotational invariance $\rightsquigarrow$ angular momentum conservation

Turning back to the central potential, since $\hat{V} \neq \hat{V}(\vartheta) \Rightarrow \hat{H} \neq \hat{H}(\vartheta)$

$\Rightarrow \hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}$ is conserved!

$\rightsquigarrow$ Theory of angular momentum

Here, we can study only a few aspects. In the position representation & spherical coordinates

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}} \doteq -i\hbar \vec{r} \times \vec{\nabla} = -i\hbar \vec{r} \times \left(e_r \frac{\partial}{\partial r} + e_\vartheta \frac{1}{r}\frac{\partial}{\partial \vartheta} + e_\varphi \frac{1}{r\sin\vartheta}\frac{\partial}{\partial \varphi} \right)$$

$$= -i\hbar r\, e_r \times \left(e_r \frac{\partial}{\partial r} + e_\vartheta \frac{1}{r}\frac{\partial}{\partial \vartheta} + e_\varphi \frac{1}{r\sin\vartheta}\frac{\partial}{\partial \varphi} \right)$$

$$= -i\hbar \left(e_\varphi \frac{\partial}{\partial \vartheta} - e_\vartheta \frac{1}{\sin\vartheta}\frac{\partial}{\partial \varphi} \right) \qquad e_i \cdot e_j = \delta_{ij}$$

$$\Rightarrow \quad \hat{\vec{L}}^2 \doteq -\hbar^2 \left[\frac{1}{\sin\vartheta}\frac{\partial}{\partial \vartheta}\left(\sin\vartheta \frac{\partial}{\partial \vartheta}\right) + \frac{1}{\sin^2\vartheta}\frac{\partial^2}{\partial \varphi^2} \right]$$

$$= -\hbar^2 r^2 \left[\nabla^2 - \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) \right]$$

where ∇^2 denotes the Laplacian in spherical coord

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{1}{\sin\vartheta}\frac{\partial}{\partial \vartheta}\left(\sin\vartheta \frac{\partial}{\partial \vartheta}\right) + \frac{1}{r^2\sin^2\vartheta}\frac{\partial^2}{\partial \varphi^2}$$

We can express the (measurable) components $\hat{L}_i$ ($i = 1,2,3$ or x,y,z) as projections in polar coordinates

$$\hat{L}_i \equiv e_i \cdot \hat{\vec{L}}$$

$$\Rightarrow \quad \hat{L}_z = -i\hbar \left(e_r \cos\vartheta - e_\vartheta \sin\vartheta \right) \cdot \left(e_\varphi \frac{\partial}{\partial \vartheta} - e_\vartheta \frac{1}{\sin\vartheta}\frac{\partial}{\partial \varphi} \right) = -i\hbar \frac{\partial}{\partial \varphi}$$

$$\hat{L}_x = -i\hbar \left(e_r \sin\vartheta\cos\varphi + e_\vartheta \cos\vartheta\cos\varphi - e_\varphi \sin\varphi \right)\left(e_\varphi \frac{\partial}{\partial \vartheta} - e_\vartheta \frac{1}{\sin\vartheta}\frac{\partial}{\partial \varphi} \right) = i\hbar \left(\sin\varphi \frac{\partial}{\partial \vartheta} + \cot\vartheta\cos\varphi \frac{\partial}{\partial \varphi} \right)$$

$$\hat{L}_y = -i\hbar \left(e_r \sin\vartheta\sin\varphi + e_\vartheta \cos\vartheta\sin\varphi + e_\varphi \cos\varphi \right)(\ldots) = i\hbar \left(-\cos\varphi \frac{\partial}{\partial \vartheta} + \cos\vartheta\sin\varphi \frac{\partial}{\partial \varphi} \right)$$

Per otra parte, using $\hat{L}_i \equiv \epsilon_{ijk}\, \hat{x}_j \hat{p}_k$

$$\Rightarrow \quad \begin{aligned} \hat{L}_x &= \hat{y}\hat{p}_z - \hat{z}\hat{p}_y \\ \hat{L}_y &= -\hat{x}\hat{p}_z + \hat{z}\hat{p}_x \\ \hat{L}_z &= -\hat{y}\hat{p}_x + \hat{x}\hat{p}_y \end{aligned}$$

we find

$$\begin{aligned} [\hat{L}_x, \hat{L}_y] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y \,,\, -\hat{x}\hat{p}_z + \hat{z}\hat{p}_x] = [\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] + [\hat{z}\hat{p}_y, \hat{x}\hat{p}_z] \\ &= [\hat{p}_z, \hat{z}]\hat{y}\hat{p}_x + [\hat{z}, \hat{p}_z]\hat{x}\hat{p}_y = -i\hbar\, \hat{y}\hat{p}_x + i\hbar\, \hat{x}\hat{p}_y \\ &= i\hbar \hat{L}_z \end{aligned}$$

$$\& \quad [\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x \quad , \quad [\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y \quad , \quad [\hat{L}_i, \hat{L}_i] = 0$$

$$\begin{aligned} \Rightarrow \quad [\hat{L}^2, \hat{L}_z] &= [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 , \hat{L}_z] = [\hat{L}_x^2 + \hat{L}_y^2, \hat{L}_z] \\ &= [\hat{L}_x^2, \hat{L}_z] + [\hat{L}_y^2, \hat{L}_z] \\ &= \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y \\ &= -i\hbar \hat{L}_x \hat{L}_y - i\hbar \hat{L}_y \hat{L}_x + i\hbar \hat{L}_y \hat{L}_x + i\hbar \hat{L}_x \hat{L}_y = 0 \end{aligned}$$

$$\& \quad [\hat{L}^2, \hat{L}_i] = 0 \qquad i = 1, 2 \quad \text{too.}$$

$\Rightarrow$ One can build simultaneous states of $\hat{L}^2$ & $\hat{L}_i$, but since $[\hat{L}_i, \hat{L}_j] \neq 0$, one <u>must choose an i</u>. We choose $\hat{z}$.

$$[\hat{L}^2, \hat{L}_z] = 0$$

$$\Rightarrow \quad \hat{L}^2\, Y(\vartheta, \varphi) = \hbar^2 \lambda\, Y(\vartheta, \varphi) \qquad 0 \le \vartheta \le \pi$$

$$\hat{L}_z\, Y(\vartheta, \varphi) = \hbar m\, Y(\vartheta, \varphi) \qquad 0 \le \varphi \le 2\pi$$

where m, λ are <u>adimensional</u> eigenvalues.

$$\Rightarrow \qquad -i\hbar \frac{\partial}{\partial\varphi} Y(\vartheta,\varphi) = \hbar m\, Y(\vartheta,\varphi)$$

$$\Rightarrow \quad Y(\vartheta,\varphi) = \Theta(\vartheta)\, e^{im\varphi}$$

Since φ & $\varphi + 2\pi$ leads to the same result (rotation by 2π)

$$\Rightarrow \qquad m \in \mathbb{Z}$$

Further, $\hat{L}^2 Y = \hbar^2 \lambda Y$ leads to

$$-\hbar^2 \left[\frac{1}{\sin\vartheta}\frac{\partial}{\partial\theta}\left(\sin\vartheta \frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\vartheta}\frac{\partial^2}{\partial\varphi^2}\right] Y(\vartheta,\varphi) = \hbar^2 \lambda\, Y(\vartheta,\varphi)$$

Replacing we get (USE full derivative $\partial \to d$)

$$\frac{1}{\sin\vartheta}\frac{\partial}{\partial\vartheta}\left(\sin\vartheta \frac{\partial}{\partial\theta}\right)\Theta - \frac{m^2}{\sin^2\vartheta}\Theta + \lambda\Theta = 0$$

Using $x = \cos\vartheta \quad \Rightarrow \quad x \in (-1,+1)$

$$\sin^2\vartheta = 1 - x^2$$

$$\frac{1}{\sin\vartheta}\frac{\partial}{\partial\theta}\left(\sin\vartheta\frac{\partial}{\partial\theta}\right)\Theta = \frac{1}{\sin\vartheta}\cancel{\frac{\partial x}{\partial\theta}}\frac{\partial}{\partial\theta}\left(\sin\vartheta \frac{\partial x}{\partial\theta}\frac{\partial}{\partial x}\right)\Theta$$

$$= \cancel{\frac{1}{\sin\vartheta}\frac{\partial x}{\partial\theta}\frac{\partial x}{\partial\theta}} = \frac{1}{\sin\vartheta}\frac{\partial}{\partial\theta}\left(-\sin^2\vartheta\frac{\partial}{\partial x}\Theta\right)$$

$$= \frac{1}{\sin\vartheta}(-2\sin\vartheta)\cos\vartheta\frac{\partial}{\partial x}\Theta - \sin\vartheta\frac{\partial x}{\partial\vartheta}\frac{\partial^2}{\partial x^2}\Theta$$

$$= -2x\frac{\partial}{\partial x}\Theta + \sin^2\vartheta\frac{\partial^2}{\partial x^2}\Theta = -2x\frac{\partial\Theta}{\partial x} + (1-x^2)\frac{\partial^2}{\partial x^2}\Theta$$

$$\Rightarrow \quad (1-x^2)\frac{\partial^2}{\partial x^2}\Theta - 2x\frac{\partial}{\partial x}\Theta + \left(\lambda - \frac{m^2}{1-x^2}\right)\Theta = 0$$

which has a regular (non-singular) solution only if

$$\lambda = l(l+1), \quad l \in \mathbb{N} \qquad -l \le m \le l$$

(see app. A3 of De La Peña).

The solutions are the well known <u>Associated Legendre Polynomials</u> which can be obtained from the Rodrigues formula

$$P_l^m(x) = \frac{(-1)^m}{2^l l!}(1-x^2)^{m/2}\frac{d^{l+m}}{dx^{l+m}}(x^2-1)^l \qquad \begin{array}{l} l = 0,1,2,\dots \\ -l \le m \le l\ , \quad m \in \mathbb{Z}\end{array}$$

satisfying $\int_{-1}^{1} P_l^m(x) P_{l'}^m(x)\,dx = \frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{ll'}$. The first polynomials are

$P_0^0 = 1$

$P_1^{+1} = -\sqrt{1-x^2}\ ,\quad P_1^0 = x\ ,\quad P_1^{-1} = -\frac{1}{2}P_1^1$

$P_2^2 = 3(1-x^2)\ ,\quad P_2^1 = -3x\sqrt{1-x^2}\ ,\quad P_2^0 = \frac{1}{2}(3x^2-1)\ ,\quad P_2^{-1} = -\frac{1}{6}P_2^1\ ,\quad P_2^{-2} = \frac{1}{24}P_2^2$

$\vdots$

We have finally found that in a system with central potential $V(r)$, the eigenfunctions of the conserved angular momentum are

error

$$Y_l^m(\vartheta,\varphi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\,\cancel{(-1)}^m P_l^m(\cos\vartheta)\,e^{im\varphi} \qquad m \ge 0$$

which (for $m \ge 0$) are called the spherical harmonics &

$$Y_l^{-m} = (-1)^m Y_l^{m\,*}$$

normalized such that

$$\int_0^{2\pi} d\varphi \int_0^{\pi} \sin\vartheta\, d\vartheta\ Y_l^{m*}(\vartheta,\varphi)\, Y_{l'}^{m'}(\vartheta,\varphi) = \delta_{ll'}\,\delta_{mm'}$$

<u>AND</u> we also found that

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{1}{\hbar^2 r^2}\hat{L}^2$$

which appears in the Schrödinger eq.

$$\left(-\frac{\hbar^2}{2m}\vec{\nabla}_r^2 + V_{int}(\vec{r})\right)\psi(\vec{r}) = E\psi(\vec{r})$$

$$\Leftrightarrow \left[-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{1}{\hbar^2 r^2}\hat{L}^2\right) + V_{int}(r)\right]\psi(\vec{r}) = E\psi(\vec{r})$$

Propose the ansatz

$$\psi(\vec{r}) = R(r)\, Y_\ell^m(\vartheta,\varphi)$$

Inserting in the Schrödinger eq. & dividing by RY_ℓ^m we obtain

$$-\frac{1}{R}\frac{\hbar^2}{2mr^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right)R + V_{int}(r) - E + \frac{\hbar^2}{2m\hbar^2}\frac{1}{r^2 Y_\ell^m}\underbrace{\hat{L}^2 Y_\ell^m}_{=\hbar^2\ell(\ell+1)Y_\ell^m} = 0$$

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{2m}{\hbar^2}\left(E - V_{int}(r)\right) - \frac{\ell(\ell+1)}{r^2}\right]R = 0$$

For historical reasons, ℓ & m are called

ℓ: orbital quantum number

m: magnetic quantum number

The great simplification of the system is evident if we propose the ansatz

$$u(r) = r\,R(r)$$

express as $\frac{d^2u}{dr^2}$

$$\Rightarrow \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}R\right) = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\left(\frac{u}{r}\right)\right) = \frac{1}{r^2}\left[u' + ru'' - u'\right] = \frac{u''}{r}$$

$$\Rightarrow \quad \frac{d^2u}{dr^2} + \left[\frac{2m}{\hbar^2}\left(E - V_{int}(r)\right) - \frac{\ell(\ell+1)}{r^2}\right]u = 0$$

or

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \underbrace{\left(V_{int}(r) + \frac{\hbar^2}{2m}\frac{\ell(\ell+1)}{r^2}\right)}_{\equiv V_{eff}(r)}u = E\,u$$

Schrödinger eq. in one dim. with $V_{eff}(r)$!

Due to the factor $\frac{1}{r^2}$ it's clear that, in general, this eq. has no sol. for $r=0$; thus, we must impose a boundary condition such as

$$\boxed{u(0) = 0}$$

which may be interpreted as $V(r\leq 0)\to\infty$ to avoid sols. $u(r)$, $r<0$.